Data Types and Variables

LEARNING OBJECTIVES

At the end of this lesson, you should be able to:

- Discuss the concept of data types and their importance in programming.
- Describe the different data types available in Java and their characteristics.
- Demonstrate how to declare and use variables to store and manipulate data.
- Apply data types and variables effectively in practical programming scenarios.

Introduction

Data types and variables are fundamental concepts in programming that allow us to represent, store, and manipulate data. They form the building blocks of any program and enable us to work with various kinds of information.

Data types define the kind of data that can be stored in a variable. Java has two main categories of data types: primitive and reference.

PRIMITIVE DATA TYPES

byte: Represents a small integer (byte smallNum = 42;).

short: Represents a short integer (short smallInt = 1000;).

int: Represents an integer (int intNum = 12345;).

long: Represents a long integer (long longNumb = 9876543210L;).

float: Represents a floating-point number (float fPNum = 3.14f;).

double: Represents a double-precision floating-point number (double dBFPNum = 2.71828;).

char: Represents a single character (char singleChar = 'A';).

boolean: Represents a true or false value (boolean isTrue = true;).

Reference Data Types

Objects: Represent instances of classes in Java.

Arrays: Represent collections of elements of the same data type.



DECLARING AND INITIALIZING VARIABLES

Variables are named storage locations used to hold values of a specific data type.

Example

```
int age; // Declaration
age = 25; // Initialization
```

Naming Conventions and Best Practices

Variable names should be meaningful and follow camelCase.

Choose descriptive names that reflect the purpose of the variable.

Avoid using reserved words as variable names.

Example

```
int studentAge;
double averageScore;
String studentName;
```

VARIABLE SCOPE AND LIFETIME

Scope defines where a variable can be accessed within the code.

Lifetime refers to the duration for which a variable exists in memory.

Example

```
int x = 10; // x is accessible in this scope
if (x > 5) {
    int y = 20; // y is accessible only within this block
}
```

MID-LESSON QUESTION

1. Write a Java program to calculate the total cost of purchasing a number of items. Each item has a price and a quantity. Calculate and display the total cost.



ANSWER :MID-LESSON QUESTION

```java
public class TotalCostCalculator {
    public static void main(String[] args) {
        // Item details: price and quantity
        double item1Price = 10.99;
        int item1Quantity = 2;

        double item2Price = 5.49;
        int item2Quantity = 3;

        double item3Price = 2.99;
        int item3Quantity = 5;

        double item4Price = 7.95;
        int item4Quantity = 1;

        double totalCost = 0.0;
```

ANSWER : MID-LESSON QUESTION

```
        double item1Cost = item1Price * item1Quantity;
        totalCost += item1Cost;
        System.out.println("Cost for Item 1: $" + item1Cost);

        double item2Cost = item2Price * item2Quantity;
        totalCost += item2Cost;
        System.out.println("Cost for Item 2: $" + item2Cost);

        double item3Cost = item3Price * item3Quantity;
        totalCost += item3Cost;
        System.out.println("Cost for Item 3: $" + item3Cost);

        double item4Cost = item4Price * item4Quantity;
        totalCost += item4Cost;
        System.out.println("Cost for Item 4: $" + item4Cost);

        System.out.println("Total Cost: $" + totalCost);
    }
}
```

SUMMARY

Summary

- Data types define the kind of data variables can store.
- Java has primitive and reference data types.
- Variables are named storage locations for data.
- Proper naming conventions and best practices enhance code readability.
- Variable scope and lifetime determine accessibility and memory usage.

FURTHER READING RESOURCES

- Sierra, K., & Bates, B. (2005). Head First Java: A Brain-Friendly Guide. " O'Reilly Media, Inc.".
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- Cosmina, I. (2018). Java for Absolute Beginners: Learn to Program the Fundamentals the Java 9+ Way. Apress.
- Schildt, H. (2003). Java™ 2: A Beginner's Guide

Thank You